

THE AI TOOLBOX · CLASS 12

Decision Trees & Random Forests

Learn models you can **read** — and why many trees beat one.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

Categories, via probabilities

The sigmoid turns a score into $P(\text{class} = 1)$; the threshold turns that into a decision.

The boundary is the model

A straight line through the feature space — move the threshold, the line slides.

Accuracy can lie

96% and zero pedestrians found. Report the confusion matrix.

Homework share: which threshold did you ship, and what did it cost you? Now notice what both models so far had in common — **they draw one straight line and hope the world cooperates.** Today's model doesn't draw lines at all. It asks questions.

Where we're going

- **0–8'** Homework share + 787 people guess the weight of an OX
- **8–18'** If-then trees: root, question, branch, leaf — and how a split gets chosen
- **18–26'** One tree overfits — watch train and test come apart
- **26–34'** Forests: why many mediocre voters beat one expert
- **34–42'** Demos: print a tree's rules in Colab · the AI rule-reader
- **42–54'** Two labs: pick the best question with Gini by hand · tree vs. forest, for real
- **54–60'** Your rule-based model, homework & wrap-up
- Quizzes & 💬 comments throughout

By the end: explain how a tree splits data, spot overfitting in one tree, and say why a forest beats it — with your own numbers.

WARM-UP · ACTIVATE

Quick check · tap an answer

"If it's a pedestrian AND closer than 10 metres, brake." Could a machine-learning model's entire logic look like that — readable, in plain language?

No — a trained model is always a wall of numbers you can't interpret

Yes: a decision tree literally is a nested set of if-then questions, and you can print every one of them

Only if a human writes the rules by hand first

Only for very small datasets under about 100 rows

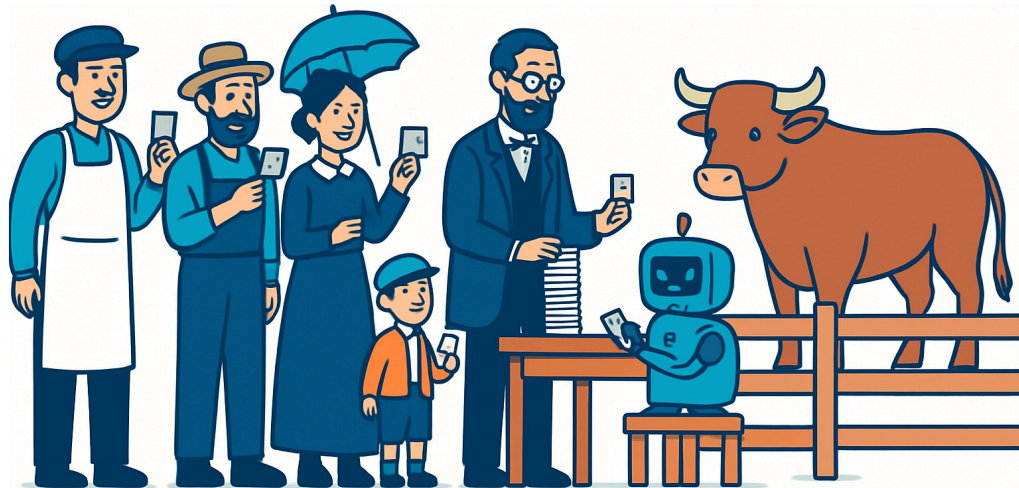
The crowd that guessed the ox

At a country fair in Plymouth in **1906**, an ox was put on display and fairgoers paid sixpence to guess its weight. **787** legible tickets came in — butchers, farmers, and people who had never touched a cow. Francis Galton — the same Galton from Class 10 — collected every ticket expecting to prove the crowd was foolish. The middle guess was **1,207 lb**. The ox weighed **1,198 lb**. The crowd was off by **0.8%**, and beat every individual expert in the room. Galton, to his credit, published it anyway.



Speak this story

Neural voice · click to play / pause



When statisticians later recomputed the **mean** of Galton's published guesses, it came to 1,197 lb — one pound *under* the true weight. That's the random forest in one anecdote: **one tree is a single fairgoer; the forest is the crowd.**

A model made of questions

A decision tree splits the data by asking one yes/no question about one feature at a time, until each group is pure enough to answer.

- **Root** — all your rows, before any question
- **Internal node** — a test on one feature: *is distance ≤ 10 m?*
- **Branch** — the yes side and the no side
- **Leaf** — no more questions: here is the answer

```
is distance_m  $\leq$  10?  
├ yes → BRAKE  
└ no  → is rel_speed  $\geq$  55?  
      ├── yes → BRAKE  
      └ no  → GO
```

KEY IDEA — trees are **interpretable**. You can follow exactly why the model decided, in words, for any single row. No other model in this course lets you do that. Two side benefits: no scaling needed, and numbers and categories mix freely.

Purity: the tree's only goal

A good question leaves the two sides more **pure** — closer to all-one-answer — than the group it started with. **Gini impurity** is the score: 0.0 = perfectly pure, 0.5 = a coin flip.

$$\text{Gini} = 1 - \sum p_i^2$$

The tree tries **every threshold on every feature**, keeps the one with the biggest drop in impurity, and then does the whole thing again on each side. That's the entire algorithm.

Worked, on our 14 rows

Root: 8 BRAKE / 6 GO → Gini **0.490**

Best question found: **distance_m ≤ 10**

→ left: 6 BRAKE / 0 GO · Gini **0.000**

→ right: 2 BRAKE / 6 GO · Gini **0.375**

Weighted impurity after: **0.214** — a gain of **0.276**

Note the tie. Two different questions score the same gain of 0.276 here — $\text{distance_m} \leq 10$ and $\text{rel_speed_kmh} \leq 42.5$. The tie-break is arbitrary, so **add two rows and you can get a completely different tree**. Remember that in twenty minutes.

Quick check · tap an answer

A node holds 20 rows: 10 BRAKE and 10 GO. A candidate question sends 10 rows left (10 BRAKE, 0 GO) and 10 rows right (0 BRAKE, 10 GO). What is the impurity after this split?

0.5 — it was 0.5 before, and splitting doesn't change the data

0.0 — both sides are perfectly pure, so the question has completely separated the classes

1.0 — the maximum, because the split is drastic

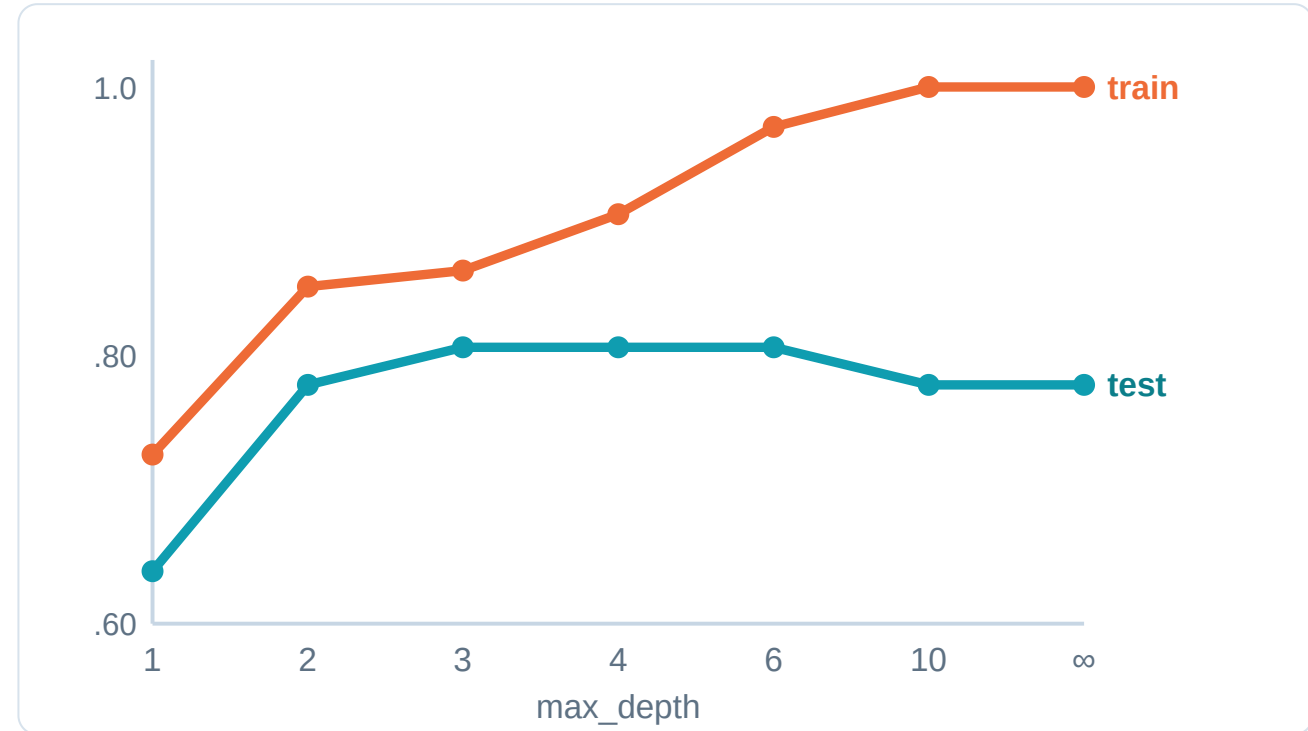
You can't tell without knowing which feature was used

One tree keeps asking until it has memorised you

Nothing stops a tree from splitting until every leaf holds one row. At that point it is **100% accurate on the training set** — and it has learned the noise along with the pattern.

- Depth 3: test 0.806 — the sweet spot
- Depth 10: train **1.000**, test **0.778**
- The gap is the overfitting, measured

This is Class 3's warning and Class 10's train/test rule, finally visible as a picture instead of a principle.



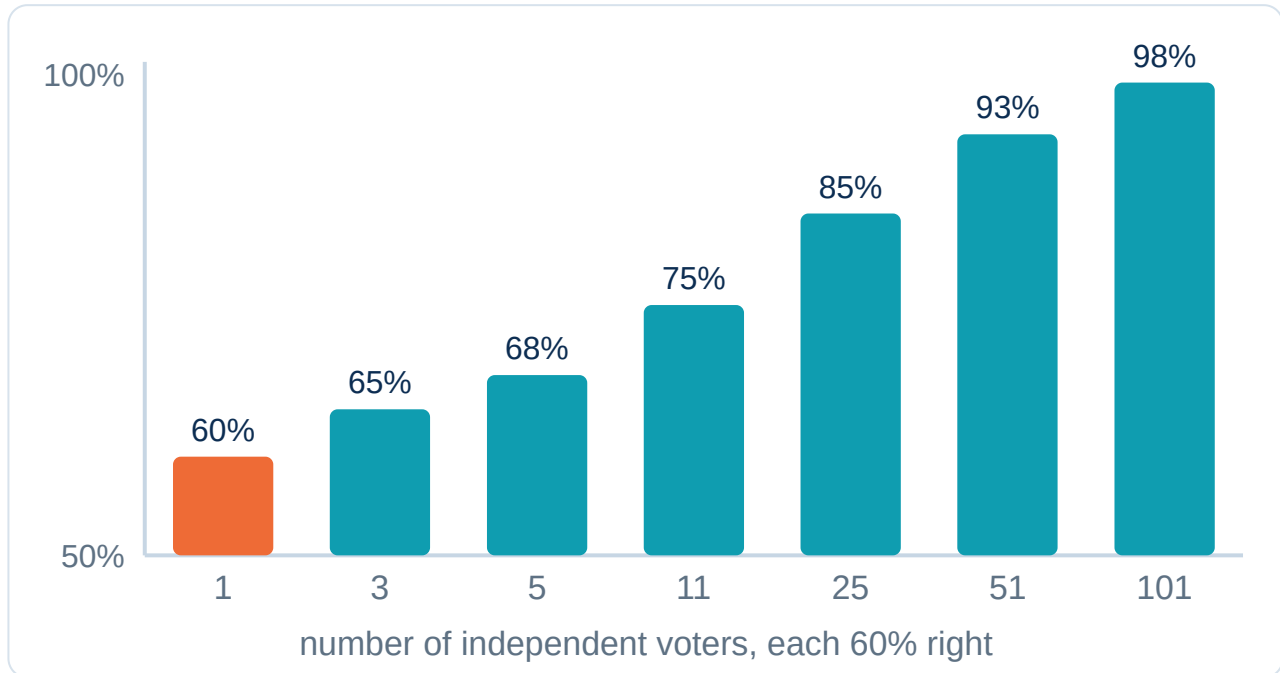
Two fixes. Pruning — stop early (max_depth, min_samples_leaf) and accept a worse training score for a better real one. Or the better fix, which is the second half of this hour: **don't build one tree.**

Many mediocre voters beat one expert

A **random forest** trains hundreds of trees, each on a different random sample of the rows *and* a random subset of the features, then lets them vote.

- Each tree is deliberately made **different**, and each is mediocre
- Their errors point in different directions and cancel
- What survives the averaging is the signal

Suppose each voter is right 60% of the time and votes independently. The chart shows the exact probability that the majority is right.



1 voter: 60%. 101 voters: 97.9%. That's Condorcet's jury theorem, and it's the ox at the fair. The catch — and it's a real one — is **independence**: real trees see overlapping data and correlate, so real forests gain much less than this chart promises. Ours went 0.778 → 0.861. Still free accuracy, for the price of readability.

Should the car brake?

The one AV decision where "the model said so" is not an acceptable answer to an investigator.

1

Features

distance_m, rel_speed_kmh, is_pedestrian

2

Tree

Its branches read like driving logic

3

Read it

"speed > 54.8 and distance ≤ 57.8 → BRAKE"

That third step is why this model still exists. After a crash, a regulator can print the tree and argue with a specific branch. A 200-tree forest scores better and can't be argued with — so safety-critical systems often ship the readable model on purpose, and keep the forest as the benchmark it has to nearly match.

DISCUSS · 5 MINUTES · WRITE A RULE

One rule you'd want a car to follow

Write a single if-then rule you'd want a self-driving car to obey. Then ask: **could a tree learn it from data?**

Make it measurable

Which columns does your rule need? If the car can't measure it, the tree can't split on it — "if the child looks like they might run" is not a feature.

Then break it

When would your rule be actively dangerous? Every rule has a situation it was never written for.

Pairs, 3 minutes — best rule and best counter-example in .

LIVE DEMO 1

Print a model's mind in Colab

Fit a shallow tree, then print its logic in plain text. Three lines, and the model explains itself.

```
You ▶ tree = DecisionTreeClassifier(max_depth=3, random_state=0).fit(X_train, y_train)
```

```
You ▶ print(export_text(tree, feature_names=FEATURES, class_names=["GO", "BRAKE"]))
```

```
You ▶ plot_tree(tree, feature_names=FEATURES, class_names=["GO","BRAKE"], filled=True) #  
the picture version
```

```
You ▶ forest = RandomForestClassifier(n_estimators=200, random_state=0).fit(X_train,  
y_train)
```

```
You ▶ print(sorted(zip(FEATURES, forest.feature_importances_), key=lambda p: -p[1]))
```



colab.research.google.com

LIVE DEMO 2

The AI rule-reader

Paste a tree's printout into an LLM and ask it to argue with the model — Class 4's mentor hat, aimed at logic instead of numbers.

You ▶ Here is my decision tree's `export_text` output. Translate each branch into one plain-English sentence a driver would understand.

You ▶ Which of these learned rules looks like it came from noise rather than real driving logic?

You ▶ My depth-10 tree scores 1.000 on train and 0.778 on test. Explain what happened, in two sentences.

You ▶ My forest says `rel_speed` matters 0.56 and `is_pedestrian` only 0.03. Give me two possible explanations – one boring, one worrying.



chatgpt.com

Pick the tree's first question yourself

No scikit-learn. Fourteen braking situations and a loop over every possible question — exactly what CART does, in about twenty lines you can read.

```
1 # 14 braking situations. We choose the tree's first question ourselves.
2 FEATURES = ["distance_m", "rel_speed_kmh", "is_pedestrian"]
3 rows = [
4     [ 8.0, 45, 1], [12.0, 30, 0], [ 5.0, 60, 1], [40.0, 25, 0], [ 6.5, 20, 0],
5     [30.0, 70, 1], [22.0, 15, 0], [ 9.0, 55, 0], [45.0, 40, 1], [ 3.0, 35, 1],
6     [18.0, 65, 1], [50.0, 20, 0], [11.0, 25, 0], [ 7.0, 80, 0],
7 ]
8 brake = [1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1]
9
10 def gini(labels):
11     """0.0 = perfectly pure (one class only). 0.5 = a 50/50 coin flip."""
12     if not labels:
```



Run



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Watch one tree overfit — then let 200 of them vote

240 braking situations with realistic 8% label noise. Sweep the depth, read the tree's rules aloud, then see what a forest buys you.

```
1 import random
2 from sklearn.tree import DecisionTreeClassifier, export_text
3 from sklearn.ensemble import RandomForestClassifier
4 from sklearn.model_selection import train_test_split
5 from sklearn.metrics import accuracy_score
6
7 # 240 braking situations, built from a physical rule + real-world messiness.
8 random.seed(12)
9 rows, labels = [], []
10 for _ in range(240):
11     distance = round(random.uniform(2, 60), 1) # metres to the object
12     rel_speed = round(random.uniform(5, 90), 1) # km/h closing speed
```

[▶ Run](#)[✨ Explain](#)[🐛 Debug](#)[⚡ Improve](#)[🤖 Mira](#)

Python loads on first Run

▶ Run the code to see the output here.

YOUR PROJECT

Build a rule-based model

Train a tree you can **read out loud**, then a forest, and report what the readability cost you.

- Pick a dataset with a category to predict — yours, or the braking data from Lab 2
- Fit a `DecisionTreeClassifier(max_depth=3)` and print its rules
- Fit a `RandomForestClassifier`
- Compare both on the **test** set

Deliverable

A tree with its printed rules, plus a forest, and their accuracy comparison.

Success looks like

You can read one learned rule aloud as a sentence, and you report both scores — including when the forest wins.

BUILD IT NOW · HANDS-ON

Shallow first, then deep, then the crowd

1. Split train/test **first** — same discipline as Class 10, no exceptions
2. Fit `max_depth=3` and print it with `export_text` — read one branch out loud to your neighbour
3. Sweep the depth: 1, 2, 3, 4, 6, 10, None. Record train AND test accuracy for each
4. Find where test accuracy stops improving — that's your tree's honest depth
5. Fit a `RandomForestClassifier(n_estimators=200)` and compare on the test set
6. One sentence: what did you give up to get the forest's extra accuracy?

[Colab](#)[scikit-learn](#)[matplotlib](#)[ChatGPT / Claude](#)

Instructor's tip: keep the tree at depth 3–4 while you're reading it. A depth-12 tree is technically printable and practically unreadable — which is exactly the trade-off this whole hour is about.

BENCHMARK · JUDGE AGAINST THIS

What finished work looks like



The readable tree

Depth 3, printed, with one branch quoted in English: *"speed above 54.8 km/h and distance under 57.8 m → BRAKE."* Test accuracy 0.806.



The overfitting evidence

A depth sweep showing train reaching 1.000 while test falls to 0.778 — your own numbers, not the slide's.



The forest, compared

200 trees → 0.861. Better, and you can no longer print why. Both numbers reported side by side.

The bar: a rule you can say out loud, a depth sweep that shows the gap opening, both models scored on the same held-out test set — and one sentence about the trade you made.

Today's vocabulary

Decision tree — a model made of nested yes/no questions about features; leaves hold the answers.

Pruning / max_depth — stopping the tree early on purpose, trading training accuracy for real accuracy.

Random forest — bagging with trees, plus a random subset of features at each split, so the trees genuinely differ.

Gini impurity — how mixed a group's labels are. 0 = pure, 0.5 = a coin flip. The thing a split minimises.

Ensemble — many models combined into one prediction. **Bagging** trains them in parallel on random samples and votes.

Interpretability — whether a human can follow why the model decided. Trees have it; forests trade it away.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your rule-based model** — printed depth-3 tree + forest + both test scores. Colab link in Google Classroom.
2. **Quote one learned rule** in plain English, and say whether you agree with it. Would you have written that rule yourself?
3. **Run the depth sweep** on your own data and report where test accuracy peaked — that number is different for everyone.

★ Stretch (optional)

Try **GradientBoostingClassifier** on the same split — the other family of ensemble, where trees are trained in sequence to fix each other's mistakes. Does it beat your forest?

[scikit-learn · ensembles](#)

Task 2 is the one that separates using a model from understanding one. A student who disagrees with a rule their own model learned — and can say why — has arrived somewhere real. Questions → 💬 on any slide.

TODAY'S LINE TO REMEMBER

"The result seems more creditable to the trustworthiness of a democratic judgment than might have been expected."

— Francis Galton, *Vox Populi*, *Nature*, 1907

He went looking for proof that crowds were foolish, found the opposite, and published it. A hundred years later we call it a random forest.



Speak this

WRAP-UP

Three things to keep

- A tree splits data with **if-then questions** — and you can read every one
- One deep tree **memorises**: train 1.000, test 0.778. The gap is the overfitting
- A **forest votes** — more accurate, less readable. That trade is a choice you make

Exit ticket: write one rule you'd expect your own tree to learn. Post it in .

Next · Class 13 — three more ways to classify the same thing: KNN, SVM, and your first neural network.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

Bagging vs. boosting

the other way to ensemble

Trees that predict numbers

regression trees

Bonus quiz

one more check

Bagging in parallel, boosting in sequence

Bagging (random forest)

Train many trees **independently** on random samples, then vote. Each tree is a full model; the crowd cancels their random errors. Reduces **variance** — the instability we watched in the tie-break.

Boosting (XGBoost, LightGBM)

Train trees **one after another**, each new one focused on the rows the previous ones got wrong. Reduces **bias** — and on messy tabular data it usually wins competitions outright.

Both are ensembles; they differ in whether the models are peers or a relay team. On the kind of spreadsheet data most research projects actually use, gradient boosting is frequently the strongest model available — and it is still, structurally, a pile of the little if-then trees you built tonight.

The same idea, aimed back at Class 10

Classification tree

Leaves hold a **class**. Splits chosen to reduce **Gini impurity** — make each side more one-answer.

Regression tree

Leaves hold a **number** — the average of the rows that landed there. Splits chosen to reduce **MSE** — make each side more one-value.

Remember Class 10's extrapolation disaster, where a straight line confidently predicted **-6.1 metres**? A regression tree cannot do that: its prediction is always an average of rows it actually saw, so it never invents a value outside its training range. It has the opposite flaw — it can only ever predict values it has seen before, in flat steps.

Quick check · tap an answer

Your decision tree scores 1.000 on training data and 0.62 on test. Your colleague suggests increasing `max_depth` so it "learns more." What actually happens?

Test accuracy improves — more depth means more learning

Nothing changes, because it already scores 1.000

Test accuracy gets worse or stays flat — a perfect training score means it is already memorising, and more depth memorises harder

The model will start underfitting